

Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-4498
Project Name	Import data using Transform Maps
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	ServiceNow built-in tools: Load Data progress page, Import Log, Transform History and the Employee List Report. This was a manual test; no external load tool such as JMeter, Locust or k6 was used.
Type of Testing	Load testing of a small batch (single user, 5 rows per load, repeated loads of the same data).
Target Module / API	Import Set load into u_employee_import, Transform Map execution (Employee Import to Employee Test, and Employee Transform for Employee Training Records), and the Employee List Report on u_employee_test.
Test Environment	Cloud hosted: ServiceNow Personal Developer Instance (https://dev447766.service-now.com), accessed through Google Chrome as administrator.
Test Date	01 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load the Excel employee sheet (5 rows) into the staging table using Load Data.	1	2 (1.844 recorded)	State Complete, Success; 5 processed, 5 inserts, 0 errors.
2	Run the Employee Transform map on import set ISET0010002.	1	Not timed	Transformation complete with Success.
3	Load the same employee data again (without Coalesce) and check the Employee Test table.	1	Not timed	Records inserted without errors; no duplicate Employee IDs expected.
4	Open the Employee List Report and the Employees by Department pie chart on the 11 loaded records.	1	Not timed	All 11 records displayed; pie chart shows the full count.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.844 s	Pass	Processing time of the single 5-row load (Figure 8). Only one run was timed.
2	Response Time (Max)	< 5 seconds	1.844 s	Pass	Same single run; the other steps were not timed.
3	Throughput (Req/sec)	>= 1 row/sec	About 2.7 rows/sec	Pass	5 rows in 1.844 s. The target is the team's own figure for a small batch.
4	Error Rate	< 1%	0%	Pass	0 errors out of 5 rows; transformation also completed with Success.
5	CPU Utilization	< 80%	Not measured	N/A	Server metrics were not recorded on the hosted instance.
6	Memory Utilization	< 80%	Not measured	N/A	Server metrics were not recorded on the hosted instance.

Step 4: Observations & Analysis

Key Findings:

The 5-row load completed in 1.844 seconds with 0 errors, and the transform of import set ISET0010002 finished with State Complete and Completion code Success. At this data volume the import and transform ran without any visible delay.

When the same data was loaded again with Coalesce set to false, the Employee Test table grew to 11 records, with Employee IDs S104082, S104085 and S104086 each appearing twice. This is a data integrity problem rather than a speed problem, and it makes report counts higher than they should be.

Bottlenecks Identified:

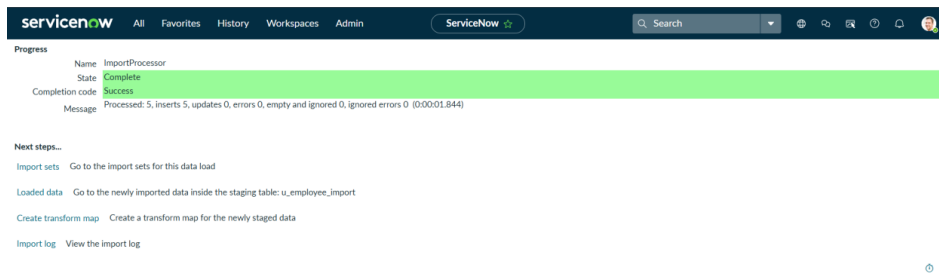
No speed bottleneck appeared with 5 to 11 rows. The limits are the small test size (large files were not tried), CPU and memory not being recorded, loading done by hand, and the Personal Developer Instance hibernating when idle.

Optimization Steps Taken:

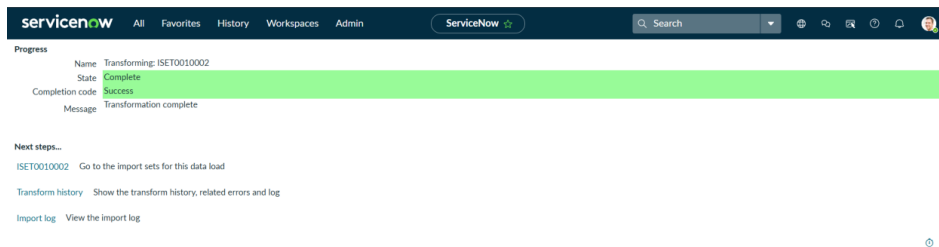
Staging tables are used so that source data is checked before it reaches the target tables, and Auto Map Matching Fields was used to create the field maps quickly. Recommended next steps are to enable Coalesce on Employee ID, index the coalesce field, split large files into batches and schedule imports outside peak hours.

Step 5: Screenshots / Evidence

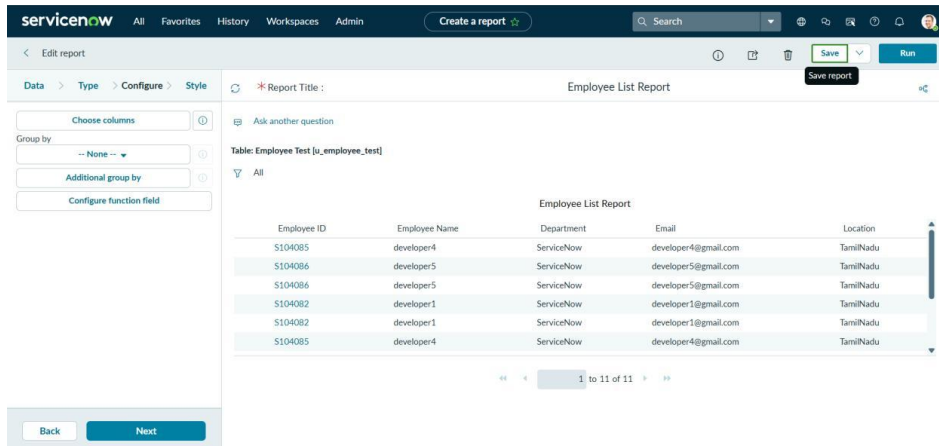
Screenshots of the ServiceNow import, transform and report results used as evidence for this test:



Screenshot 1: Load Data result - Processed 5, inserts 5, errors 0 (1.844 sec)



Screenshot 2: Transformation of import set ISET0010002 - Complete, Success



Screenshot 3: Employee List Report - 1 to 11 of 11 records